

Deep Learning Discovery of NOD2 Binders for R702W Crohn's Variant

FEP Validation of Mutation-Sensitive Binding • Tanmay Vasudeva • Texas Virtual Academy at Hallsville • ISEF 2026

INTRODUCTION

NOD2 is an intracellular pattern recognition receptor detecting bacterial muramyl dipeptide (MDP). Upon ligand binding, NOD2 activates NF-κB inflammatory signaling via RIPK2 kinase recruitment.

NOD2 mutations are the strongest genetic risk factors for Crohn's disease, conferring up to 40-fold increased risk. Three variants cause >80% of cases: R702W (~10% of patients), G908R, and L1007fs.

The R702W mutation substitutes charged arginine with neutral tryptophan in the HD2 domain, located ~80Å from the LRR binding pocket—suggesting allosteric effects.

No drugs currently target NOD2, representing significant unmet therapeutic need.

RESEARCH OBJECTIVES

- Develop computational pipeline integrating deep learning docking (GNINA), ML classification, and physics-based FEP validation
- Identify compounds that bind NOD2 with drug-like properties
- Quantify how R702W mutation affects binding affinity
- Characterize conformational dynamics differences between WT and R702W

NOD2 PROTEIN STRUCTURE

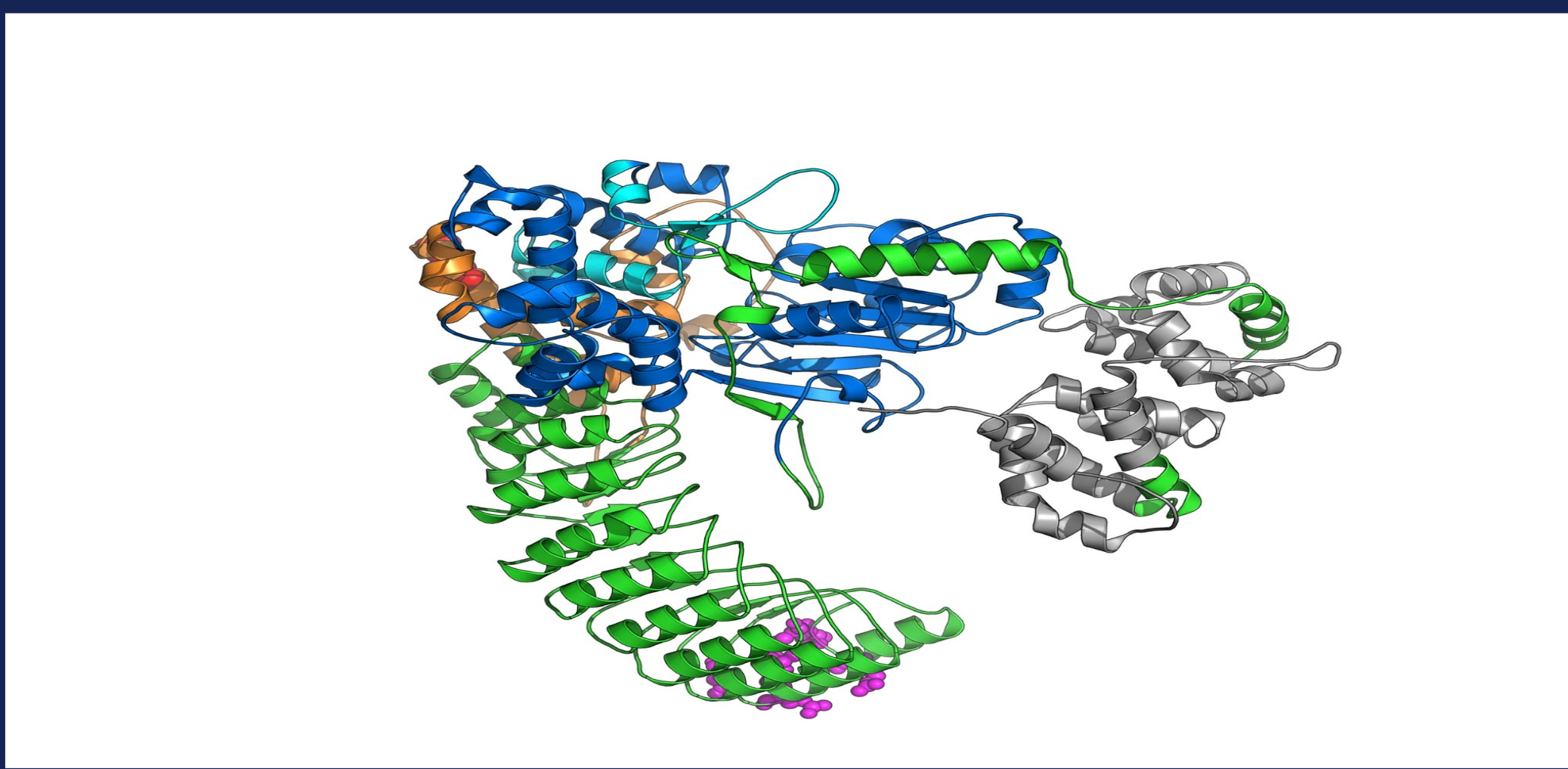


Fig 1. NOD2 domains: CARD (gray), NBD (blue), HD2/R702W (orange), LRR (green), Magenta = binding site. 80Å separates R702W from pocket.

ALLOSTERIC DISTANCE: 80Å

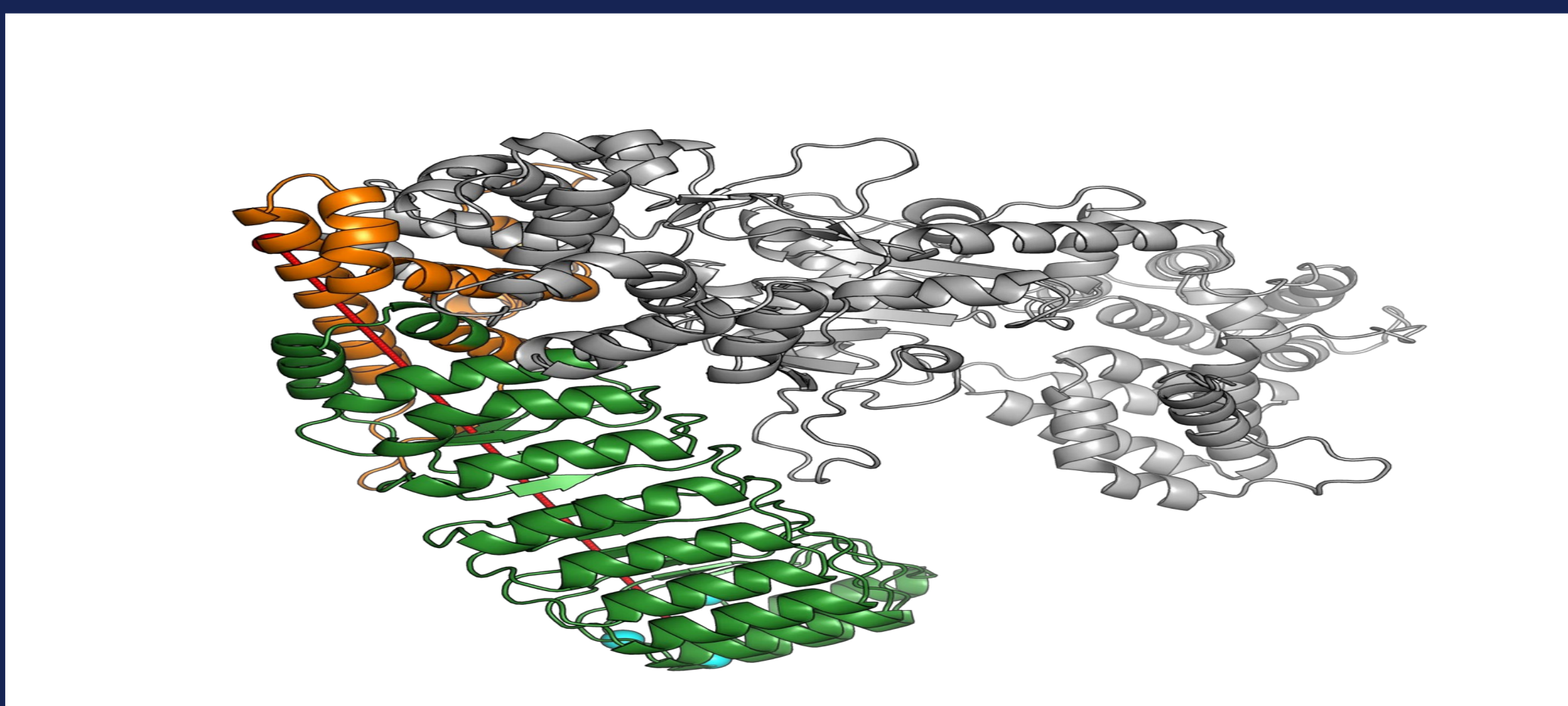
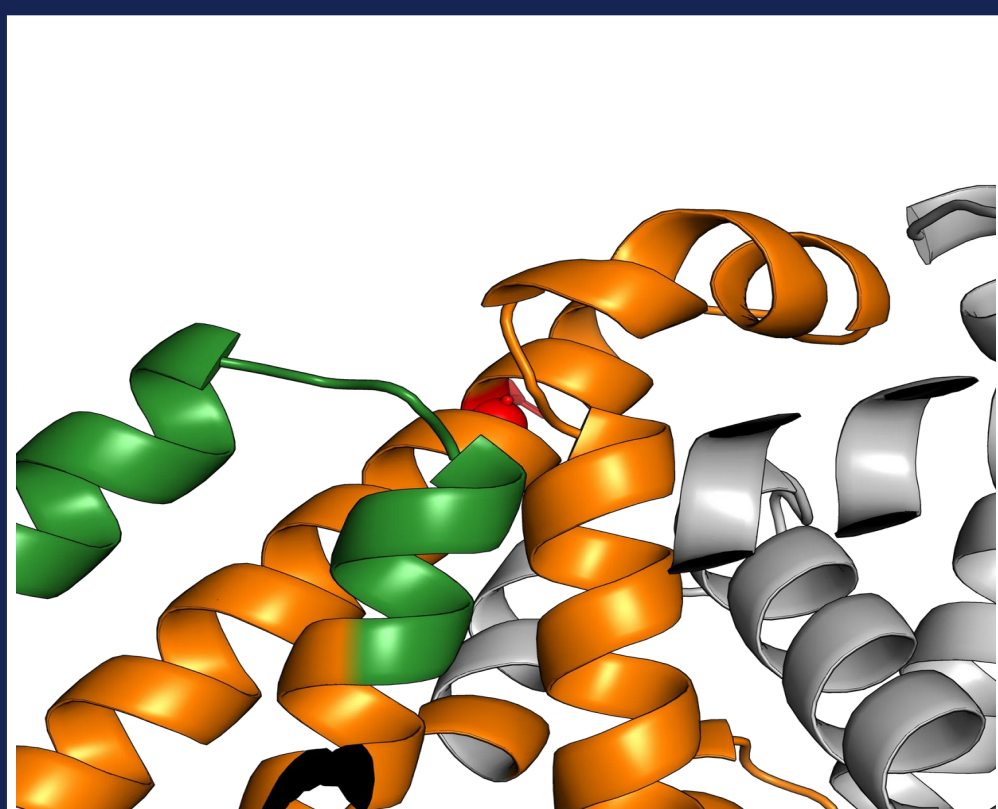


Fig 2. Red dashed line shows 80Å distance from R702W (orange) to binding pocket (cyan). Mutation effects must propagate allosterically.

R702W MUTATION SITE



- Arg → Trp: +1 charge lost
- Steric: MW 174 → 204
- H-bond donors lost
- Disrupts HD2 packing

Fig 3. HD2 helices (orange) with R702W site (red sphere).

COMPOUND SCREENING FUNNEL

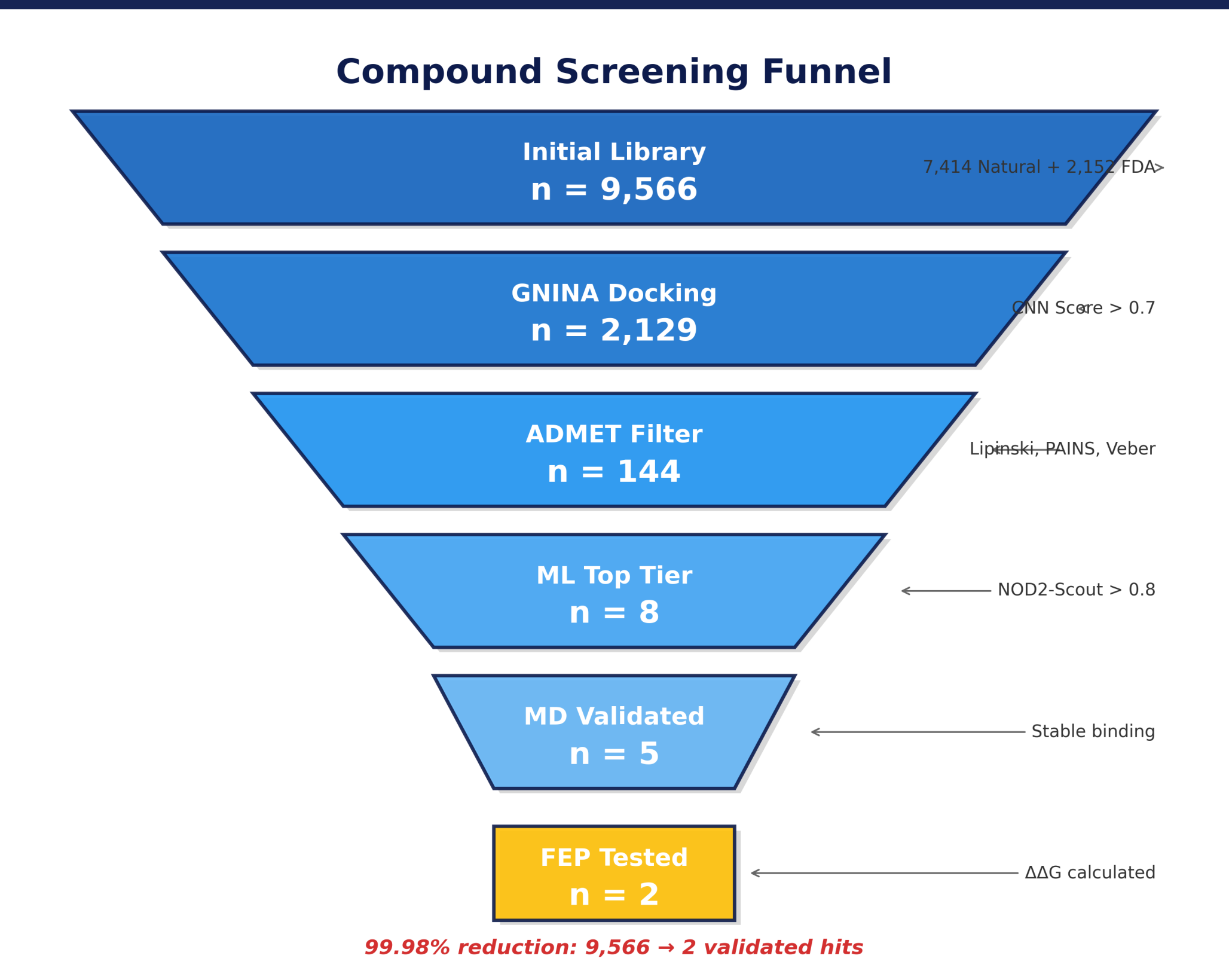


Fig 4. Multi-stage filtering: 9,566 → 2 validated hits (99.98% reduction)

ADMET & DRUG-LIKENESS FILTERING

Tool: RDKit-based offline pipeline (reproducible, no API dependency)

Filters Applied:

- Lipinski Rule of 5: MW<500, LogP<5, HBD<5, HBA<10
- Veber: Rotatable bonds ≤10, TPSA ≤140Å²
- PAINS: Pan-assay interference compounds removed
- Brenk: Structural alerts (max 2 allowed)
- BOILED-Egg: BBB/GI permeability prediction

Results: 148/150 candidates passed (98.7%), 2 failed on Brenk alerts

NOD2-SCOUT ML CLASSIFIER

Architecture: XGBoost gradient boosting classifier
Features: GNINA CNN score, Vina ΔG, MW, LogP, TPSA, HBD, HBA, rotatable bonds

Validation: 5-fold scaffold-split cross-validation

- Prevents chemotype memorization
- Tests generalization to novel scaffolds

Performance: AUC = 0.85 - 0.93 across folds

Threshold: Score > 0.8 for top-tier hits

Top Hits: Zuranolone (0.868), Desoximetasonone (0.845), Febuxostat, Budesonide

MOLECULAR DYNAMICS PROTOCOL

Software: OpenMM 8.0 (CUDA GPU acceleration)
Force Field: AMBER14 (protein) + OpenFF 2.1.0 (ligands)
Water: TIP3P-FB, 150 mM NaCl, periodic boundary
System: ~220,000 atoms, 13.2 × 13.2 × 13.2 nm box
Ensemble: NPT (310K, 1 atm), Langevin thermostat
Timestep: 2 fs with SHAKE constraints
Simulations: 520 ns total across 14 independent runs
• WT apo (n=4), R702W apo (n=3)
• WT+ligand (n=4), R702W+ligand (n=3)
Compute: ~50-60 GPU-hours (RTX 4090, 323 ns/day)

FEP CALCULATION PROTOCOL

Method: Alchemical Free Energy Perturbation (gold standard)
Thermodynamic Cycle: ΔΔG = ΔG(R702W) - ΔG(WT)

Lambda Windows: 120 total per ligand

- 80 windows: protein-ligand complex
- 40 windows: ligand in solvent (reference)

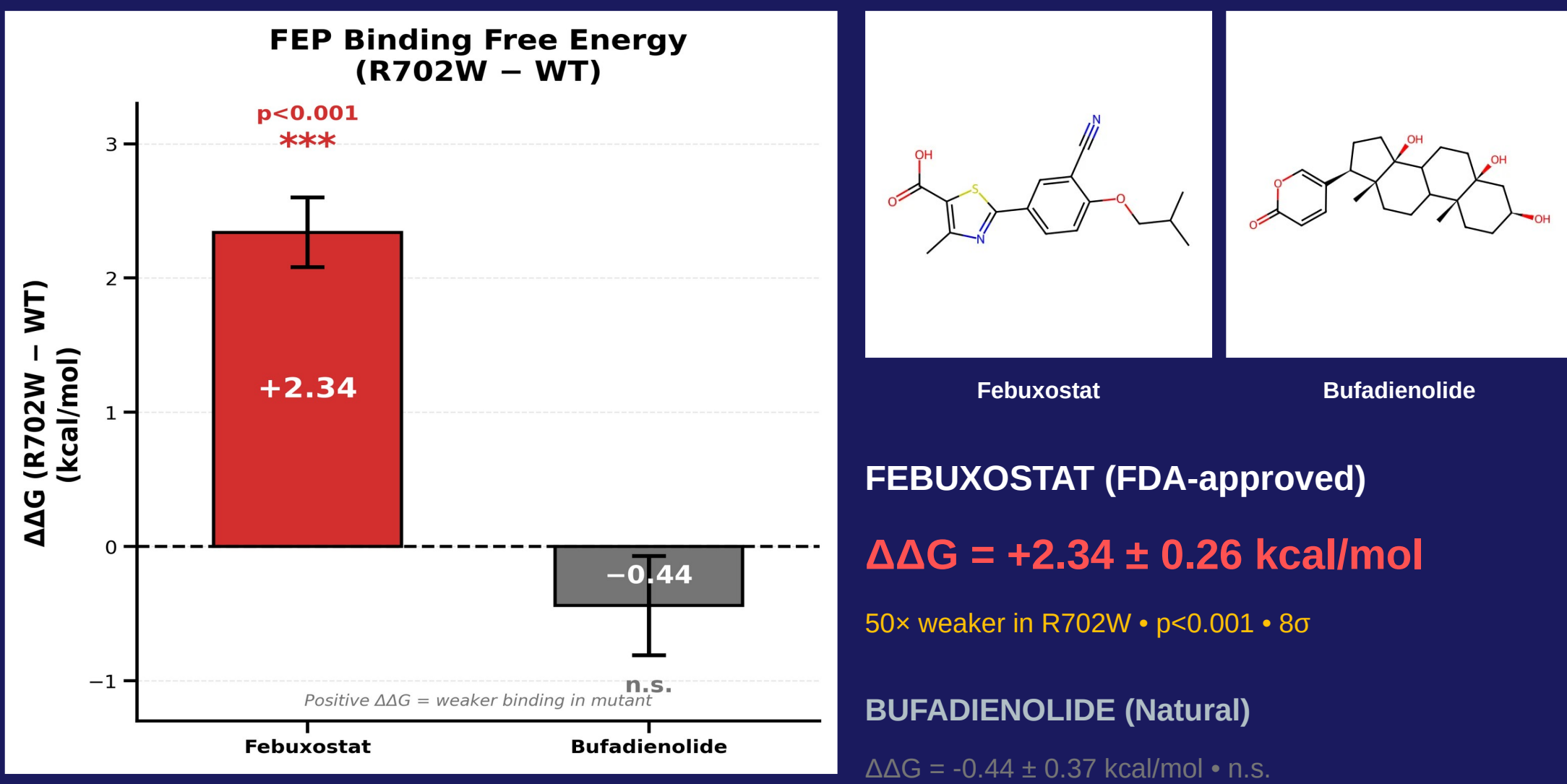
Restraints: Boresch orientational restraints

Sampling: 5 ns equilibration + 10 ns production/window

Analysis: MBAR (pymbar) with bootstrap error estimation

Overlap: Minimum λ-overlap 0.04-0.13 (good convergence)

★ KEY RESULT: FEP BINDING FREE ENERGY ★



INTERPRETATION: Mutation effects are LIGAND-DEPENDENT. Different chemotypes show different sensitivity—critical for precision medicine.

Raw ΔG Values (kcal/mol):

Ligand	WT ΔG	R702W ΔG	ΔΔG
Febuxostat	-10.36±0.18	-8.02±0.19	+2.34±0.26
Bufadienolide	-15.22±0.26	-15.66±0.26	-0.44±0.37

Fig 5. FEP ΔΔG comparison. Positive = weaker binding in mutant.

H-bond Analysis (avg ± SD):

Febuxostat: 2.02±1.78 | Bufadienolide: 1.99±1.74 | Decoy: 0.15±0.60

LIGAND BINDING IN LRR POCKET

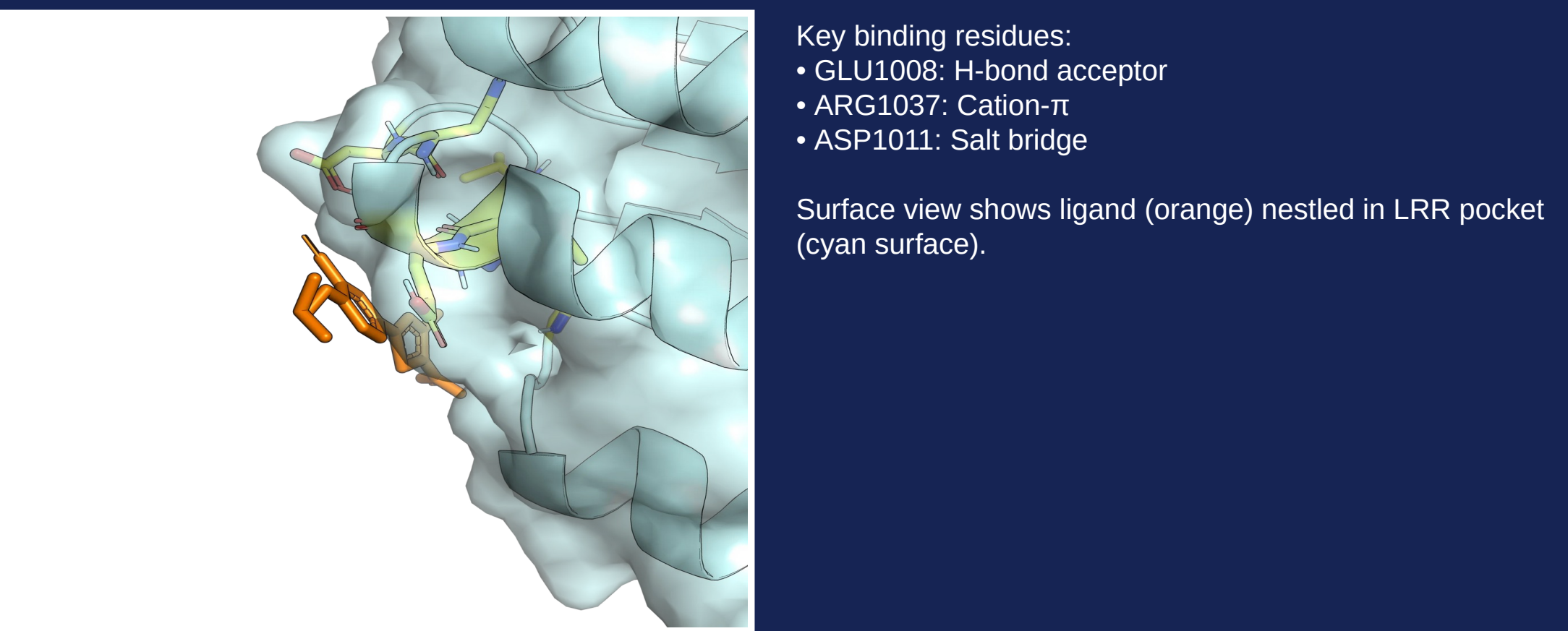


Fig 6. Binding pocket surface with ligand (orange sticks).

APO MD: RMSD ANALYSIS

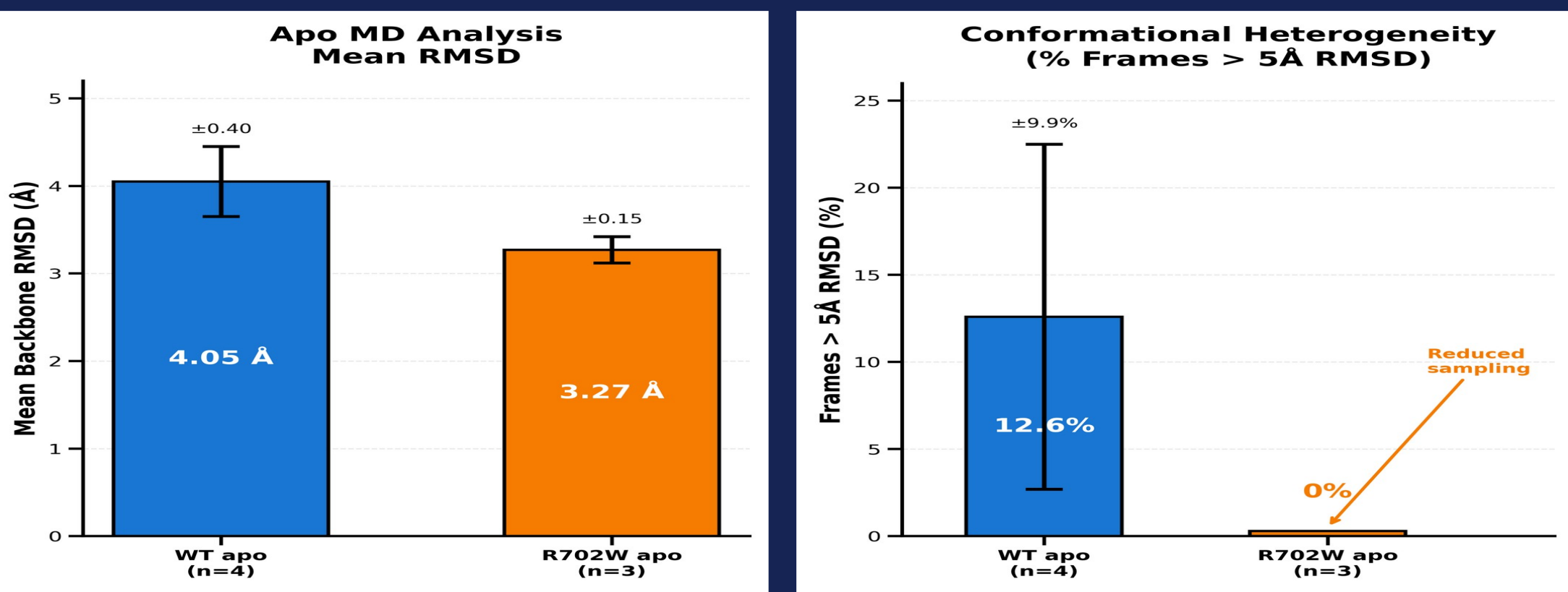


Fig 7-8. Mean RMSD and conformational heterogeneity (% >5Å). Error bars = SEM.

MD REPLICATE DATA (RMSD Å)

System	Rep1	Rep2	Rep3	Rep4	Mean	%>5Å
WT apo	3.69	5.12	3.28	4.11	4.05	12.6%
R702W apo	3.56	3.15	3.09	-	3.27	0%
WT+Feb	3.94	2.85	-	-	3.39	0%
R702W+Feb	2.95	2.99	3.15	-	3.03	0%
WT+Buf	4.05	-	3.29	-	3.67	0%
R702W+Buf	3.13	3.30	3.66	-	3.36	0%

PRECISION MEDICINE IMPLICATIONS

This work demonstrates NOD2-targeting therapeutics may need genotype matching:

- R702W patients (~10%): Febuxostat-class shows 50× reduced efficacy → use Bufadienolide-class alternatives
- WT patients: Both compound classes effective

Genotype-guided drug selection = precision medicine

CONCLUSIONS

- Developed integrated computational pipeline screening 9,566 compounds through GNINA docking, ML classification (AUC 0.85-0.93, scaffold-split CV), and ADMET filtering
- FEP provides gold-standard validation: Febuxostat binding is 50× weaker in R702W variant (ΔΔG = +2.34 kcal/mol, p < 0.001, 8σ significance)
- Mutation effects are LIGAND-DEPENDENT: Bufadienolide binding unaffected (ΔΔG = -0.44, n.s.), demonstrating compound-specific mutation sensitivity
- Apo MD reveals R702W reduces conformational heterogeneity (0% vs 12.6% frames >5Å RMSD), suggesting mutation "rigidifies" protein
- Results support ALLOSTERIC mechanism: mutation 80Å from binding site affects dynamics and binding through long-range conformational coupling
- Established framework for mutation-aware drug discovery applicable to other NOD2 variants (G908R, L1007fs)

ELECTROSTATIC SURFACE

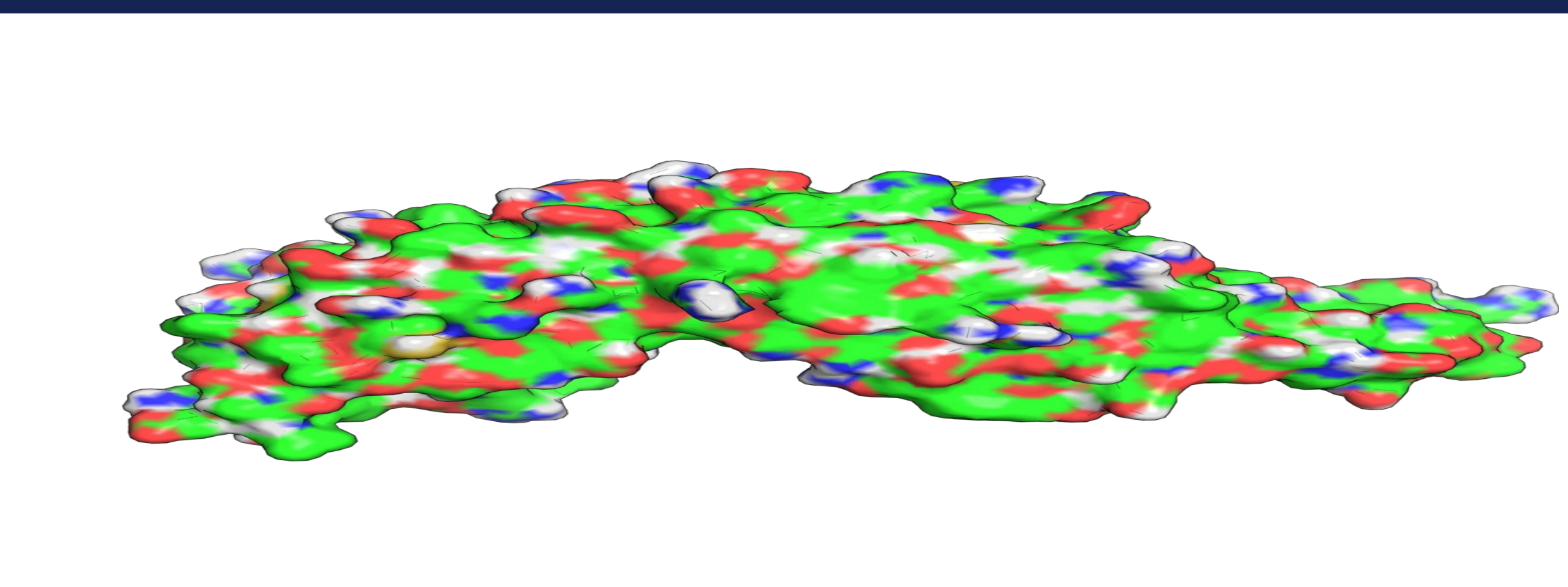


Fig 9. LRR domain electrostatic surface. Blue=positive, red=negative, green=hydrophobic.

LIMITATIONS

- FEP performed on 2 ligands; broader chemotype coverage needed
- Apo MD only; holo (ligand-bound) dynamics ongoing
- AlphaFold2 structure used; experimental validation needed
- Computational predictions require SPR/ITC confirmation
- Single mutation studied; G908R/L1007fs pending

FUTURE DIRECTIONS

- SPR/ITC binding assays for experimental validation
- HDX-MS to confirm conformational differences
- Extend FEP to G908R and L1007fs mutations
- Screen for mutation-INSENSITIVE NOD2 binders
- NF-κB reporter assays for functional validation
- Collaborate with medicinal chemists for lead optimization

COMPUTATIONAL SCALE

520 ns MD • 220,000 atoms • 120 FEP λ-windows • 50-60 GPU-hours • 9,566 compounds screened